

Quantum Phynance

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Foreword

Thanks to Alex Zecevic for letting me sit in on his quantum computing course, which motivated me to work on this project, after Brian Granger introduced me to quantum computing.

More to come.

Chapter 1

What does quantum computing mean for finance?

A SURVEY OF OPPORTUNITIES, ALGORITHMS, AND THE ROAD AHEAD

Finance has always been hungry for computation. From the Black–Scholes formula to modern Monte Carlo risk engines, the history of quantitative finance is largely a story of harnessing more computing power to solve harder problems. Quantum computing represents a potential step-change in that story—not a modest acceleration, but a qualitatively different kind of computation that may unlock problems currently beyond practical reach.

This chapter surveys what quantum computing means for finance practitioners, theorists, and regulators. We draw on a growing body of literature spanning quantum algorithms, financial applications, and empirical demonstrations on real quantum hardware. Our aim is to give the reader a clear-eyed picture: what quantum advantage means and where it genuinely applies, what the current state of hardware permits, and what the promising directions are for the decade ahead.

1.1 *Why quantum computing, and why now?*

Classical computers process information as bits—binary digits that are either 0 or 1. Quantum computers exploit the principles of quantum mechanics to process qubits, which can exist in superpositions of 0 and 1 simultaneously. Combined with entanglement—correlations between qubits with no classical analogue—and interference (the ability to amplify correct answers and suppress wrong ones), quantum computers can in principle solve certain classes of problems exponentially faster than any classical machine.

The key word is *certain*. Quantum speedups are not universal. They arise for specific algorithmic structures: unstructured search (Grover’s algorithm gives a quadratic speedup), linear systems (the HHL algorithm), quantum simulation of physical systems, and—critically for finance—sampling and Monte Carlo integration. Most financial bottlenecks fall squarely into the problems where quantum advantage has been theoretically established [[Egger et al., 2020](#); [Herman et al., 2023](#);

[Orús et al., 2019](#)].

Why now? The answer is Preskill’s NISQ era [[Preskill, 2018](#)]. Noisy Intermediate-Scale Quantum (NISQ) devices—quantum processors with 50 to a few hundred qubits, subject to noise and without full error correction—are already available through cloud platforms from IBM, Google, and others. These machines cannot yet tackle large-scale financial problems with proven quantum advantage, but they are large enough to run proof-of-concept demonstrations, to benchmark quantum algorithms against classical baselines, and to identify where quantum techniques will first become practically relevant. The financial sector has accordingly begun sustained investment in quantum research [[Auer et al., 2024](#); [Soller et al.; Soller](#)].

1.2 *The landscape of quantum finance*

The literature on quantum computing for finance has expanded rapidly. Comprehensive surveys [[Herman et al., 2022, 2023](#); [Orús et al., 2019](#); [Egger et al., 2020](#); [Bunescu and Vârtei, 2024](#); [Gong et al., 2026](#); [Zhou, 2026](#)] identify three main domains where quantum techniques offer the strongest theoretical case for advantage: simulation and sampling (quantum Monte Carlo), optimization (portfolio construction, resource allocation), and machine learning (pattern recognition, prediction). These three pillars map directly onto the core computational problems of the financial industry.

Remark 1.1. *The survey by [Herman et al. \[2023\]](#) in Nature Reviews Physics provides an especially clear roadmap. It is deliberately addressed to physicists—describing both the classical financial techniques and the quantum alternatives—and is a recommended starting point for readers new to the field.*

Alongside these opportunities, quantum computing introduces a significant risk: the ability of future quantum computers to break widely-used public-key cryptography. Financial systems rely on cryptographic protocols to secure transactions, authenticate identities, and protect sensitive data. The harvest now, decrypt later threat—in which adversaries collect encrypted data today to decrypt it once quantum computers arrive—means that post-quantum security is already a strategic necessity, not a distant concern [[Auer et al., 2024](#); [Gong et al., 2026](#)].

We treat these topics in turn.

1.3 Quantum Monte Carlo and derivative pricing

Monte Carlo simulation is the backbone of computational finance. Pricing over-the-counter derivatives, computing value-at-risk (VaR), running scenario analyses for stress testing, and calibrating stochastic models all rely on drawing large numbers of random samples and averaging a payoff or loss function. Classical Monte Carlo converges at a rate proportional

to $1/\sqrt{N}$, where N is the number of samples. Doubling precision requires quadrupling the sample count—an expensive proposition for complex, high-dimensional problems.

Quantum amplitude estimation [Egger et al., 2020], which underlies quantum Monte Carlo, achieves a quadratic speedup: it achieves the same precision with $\sqrt{N}$ samples, or equivalently, it squares the convergence rate. For financial applications that spend significant compute budget on Monte Carlo, this is potentially transformative. The catch is that quantum amplitude estimation requires coherent quantum circuits of substantial depth, which is difficult on current NISQ hardware.

Stamatopoulos et al. [2020] demonstrated a complete methodology for pricing options on a gate-based quantum computer. Using quantum amplitude estimation, they priced European calls and Asian options, constructing quantum circuits that encode the payoff distribution and estimate the expected payoff directly. The method provides a quadratic speedup over classical Monte Carlo in the number of oracle calls.

For interest-rate derivatives, Martin et al. [2021] adapted the quantum principal component analysis algorithm to the Heath–Jarrow–Morton framework. The HJM model describes forward rate dynamics through volatility factors; in its multi-factor version, the number of stochastic factors drives computational cost. Quantum principal component analysis can reduce the effective dimensionality, and the authors demon-

strated this on IBM's five-qubit quantum computer for 2×2 and 3×3 covariance matrices estimated from historical data.

[Chen et al. \[2025\]](#) provided a rigorous complexity analysis of a quantum Monte Carlo algorithm for high-dimensional Black–Scholes PDEs. For payoff functions that are continuous and piecewise affine (covering most practical payoffs), the computational complexity is bounded polynomially in both the dimension d of the PDE and the reciprocal of the prescribed accuracy ε . This establishes that quantum Monte Carlo can escape the curse of dimensionality for a broad class of option pricing problems.

[Matsakos and Nield \[2024\]](#) applied quantum Monte Carlo to financial risk analytics at a broader scale, addressing scenario generation for equity, rate, and credit risk factors simultaneously. Their work targets the computationally expensive risk measurement workflows used in regulatory capital calculations and risk management, where hundreds of thousands of Monte Carlo paths are routinely needed.

[Blanchet et al. \[2025\]](#) provide a particularly accessible bridge between classical stochastic simulation and its quantum counterpart. Starting from Grover's algorithm for unstructured search, they build intuition through amplitude estimation toward Monte Carlo integration in finance, with hands-on Python/Qiskit implementations.

1.4 *Portfolio optimization*

Portfolio optimization is one of the most natural applications of quantum computing to finance. Markowitz mean–variance optimization is a quadratic program with linear constraints. At the scale of institutional portfolios—thousands of assets, cardinality constraints, transaction cost penalties, regulatory requirements—the problem becomes combinatorial, entering the territory of NP-hard problems for which quantum annealers and variational quantum algorithms offer potential advantages.

The standard quantum approach maps portfolio selection to a Quadratic Unconstrained Binary Optimization (QUBO) problem [Orús et al., 2019; Egger et al., 2020]. Each asset is represented by a binary variable indicating inclusion or exclusion. The objective and constraints are encoded in a Hamiltonian, and the ground state of that Hamiltonian corresponds to the optimal portfolio. Two main quantum approaches then apply: quantum annealing (used by D-Wave hardware) and QAOA (Quantum Approximate Optimization Algorithm), a gate-based variational method.

Morapakula et al. [2026] present an end-to-end portfolio optimization pipeline combining continuous mean-variance and Sharpe-ratio formulations with a QUBO-based discrete asset selection stage solved on D-Wave’s hybrid quantum annealing solver, followed by classical convex optimization for portfolio weights and quarterly rebalancing. Tested on Indian equity

markets, the hybrid pipeline achieves competitive returns relative to benchmark indices. They are careful to frame the contribution as demonstrating feasibility and integration rather than claiming quantum advantage—an important distinction at the current stage of hardware maturity.

[Thomassin et al. \[2025\]](#) address a fundamental challenge in quantum portfolio optimization: how to handle inequality constraints within a QAOA framework. Their approach embeds slack variables directly into the problem Hamiltonian as ancilla qubits, converting inequality constraints into a QUBO formulation. This slack-ancilla scheme enables QAOA to find the optimal portfolio consistently in simulation, whereas standard penalty-based QAOA fails. They also derive a quantum uncertainty principle for the simultaneous precision of portfolio risk and return—a striking formal analogy between quantum mechanics and financial optimization.

[Zaman et al. \[2024\]](#) propose a scalable framework, PO-QA, for systematically studying how quantum circuit parameters (rotation blocks, repetition depth, entanglement patterns) affect QAOA and VQE (Variational Quantum Eigensolver) performance on portfolio optimization. Convergence to the classical eigensolver solution is used as the benchmark, providing practical guidance on circuit design for finance. The framework was accepted at the 2024 IEEE International Conference on Quantum Computing and Engineering.

[Chou et al. \[2026\]](#) take a different approach: quantum-in-

spired algorithms, classical algorithms that borrow ideas from quantum mechanics (tensor networks, quantum-like state evolution) without requiring quantum hardware. Applied to G7 currency portfolio management, they demonstrate that quantum-inspired optimization can already outperform standard classical methods on real financial data, offering a near-term path to value even before fault-tolerant quantum computers arrive.

1.5 *Risk management and credit risk*

Risk management encompasses a broad set of computational challenges: credit risk modeling, value-at-risk estimation, conditional value-at-risk, counterparty exposure, and stress testing. Many of these reduce to tail-probability estimation problems, which are notoriously expensive classically because the rare events that matter most require enormous simulation samples.

[Dri et al. \[2022\]](#) address credit risk analysis directly. A quantum algorithm for credit risk with quadratic speedup over classical methods had been established in prior work; they extend it to handle more realistic credit models. Their contributions include: a richer default probability model that accounts for multiple systemic risk factors (not just a single factor), and a flexible loss given default parameter that accepts real-valued inputs rather than integers. The latter is critical for using real financial data in benchmarking. While the enhanced circuits are deeper and wider than the original,

they map a path toward a practically useful quantum credit risk tool as hardware matures.

[Matsakos and Nield \[2024\]](#) address the same domain from a simulation angle. Their quantum Monte Carlo framework for financial risk analytics generates scenarios jointly for equity prices, interest rates, and credit spreads, enabling holistic risk measurement. This is relevant for derivative counterparty risk, regulatory stress testing, and internal model validation.

[Zhou \[2026\]](#) review the evidence that quantum Monte Carlo algorithms can reduce sample-size requirements by up to a factor of four compared to classical methods, with applications specifically to VaR and CVaR. They also survey the challenges: scalability, data quality, integration with legacy classical systems, and regulatory compliance—all of which must be solved before quantum risk tools can be deployed in production.

1.6 *Quantum machine learning in finance*

Machine learning has transformed quantitative finance over the past decade, and quantum machine learning (QML) promises to extend that transformation by encoding data in quantum states that classical computers cannot efficiently represent [[Biamonte et al., 2017](#)]. The key intuition is that quantum systems naturally generate probability distributions that are hard to sample classically; if financial data has a similar complex structure, quantum models may learn it more

efficiently.

[Biamonte et al. \[2017\]](#) provide the foundational survey of QML. They identify building blocks including quantum principal component analysis, quantum support vector machines, and quantum neural networks, all of which have potential speedups over their classical counterparts under specific conditions on the data-loading problem. The caveat—sometimes called the input problem—is that efficiently loading classical data into quantum states is itself non-trivial and may limit practical speedups.

In finance, QML applications include return prediction, volatility forecasting, fraud detection, and portfolio construction using quantum-enhanced feature spaces. [Herman et al. \[2023\]](#) provide a careful assessment: while the theoretical case for QML speedups is strong, the practical advantage on financial data depends heavily on the structure of that data and the efficiency of quantum data encoding.

[Samal \[2024\]](#) focus specifically on the intersection of quantum computing and Bayesian machine learning in finance. Bayesian methods are natural for financial modeling—they handle uncertainty explicitly and update beliefs as new data arrives—but Bayesian inference is computationally expensive. Quantum approaches to Bayesian inference may offer speedups for posterior sampling, relevant to risk model calibration and regime detection.

1.7 *Algorithmic trading and market microstructure*

The use of quantum computing in trading represents one of the most commercially sensitive application areas. [Ciceri et al. \[2025\]](#) provide a rare empirical demonstration using real production-scale data. They study fill probability estimation for institutional algorithmic bond trading—the problem of predicting, for a given trade inquiry in the corporate bond market, the probability that the trade will actually execute at the quoted price.

Using IBM Heron quantum processors, they apply quantum feature transformations to intraday trade event data, then feed these transformed features into classical machine learning models. The striking finding is a relative gain of up to 34% in out-of-sample test scores for models with access to quantum hardware-transformed data compared to models using either the original data or noiseless quantum simulation. They conjecture that the inherent noise in current quantum hardware contributes to this improvement—the hardware noise acts as a form of stochastic regularization that improves generalization. This empirical result is both intriguing and controversial, and invites further study.

[Huang](#) provides a hands-on introduction to quantum finance through the lens of financial technology education, covering amplitude estimation, option pricing circuits, and portfolio optimization with Qiskit.

1.8 *Post-quantum cryptography and financial security*

The threat quantum computing poses to financial security is distinct from the opportunity it offers for financial computation. Shor's algorithm can factor large integers and solve the discrete logarithm problem in polynomial time, breaking RSA and elliptic-curve cryptography—the protocols that secure interbank settlements, electronic payments, digital signatures on securities, and encrypted communications throughout the financial system.

This threat is not immediate: Shor's algorithm requires thousands of error-corrected logical qubits, which will take years or decades to achieve. However, the *harvest now, decrypt later* attack vector is operational today. Adversaries can record encrypted financial communications now and decrypt them once powerful quantum computers arrive. For data that must remain confidential for years—strategic positions, client information, regulatory filings—this is an active risk.

[Auer et al. \[2024\]](#), writing for the Bank for International Settlements, lay out both the opportunities and risks of quantum computing for the financial system. They note that central banks, including the Banque de France, Deutsche Bundesbank, and the BIS Innovation Hub, have launched Project Leap to address post-quantum migration. The National Institute of Standards and Technology (NIST) has standardized post-quantum cryptographic algorithms, and financial regu-

lators are increasingly requiring institutions to assess their cryptographic exposure.

Gong et al. [2026] make the point directly: “Post-quantum cryptography is already strategically necessary because financial infrastructures must migrate before fault-tolerant attacks arrive.” The migration challenge is substantial—it requires identifying all cryptographic dependencies across complex IT infrastructure, updating protocols, and coordinating with counterparties and market infrastructure operators.

1.9 *Current hardware, limitations, and the NISQ constraint*

Every practical claim about quantum advantage in finance must be assessed against the reality of current hardware. NISQ devices suffer from:

- **Qubit noise.** Gate errors, measurement errors, and decoherence limit the depth of circuits that can execute reliably. Typical two-qubit gate error rates are on the order of 0.1%–1%, which means circuits with thousands of gates accumulate significant error.
- **Limited qubit count.** Current devices have tens to a few hundred qubits. Financial problems of practical size—pricing a realistic exotic option, optimizing a large portfolio—require far more qubits than are available today.

- **Connectivity constraints.** Qubits are not all directly connected; two-qubit gates can only be applied to physically adjacent qubits, requiring costly SWAP operations for non-adjacent interactions.
- **No quantum memory.** Unlike classical RAM, quantum states cannot be easily stored and retrieved. All computation must happen within a single coherence time.

[Preskill \[2018\]](#) was prescient in describing this era: NISQ devices “will be useful tools for exploring many-body quantum physics, and may have other useful applications, but the 100-qubit quantum computer will not change the world right away.” Seven years later, that assessment remains accurate. The 1000+ qubit devices now available are still far from the millions of physical qubits needed for fault-tolerant operation at the scale required for transformative financial applications.

The implication is that near-term quantum finance must be pursued through hybrid quantum-classical workflows [[Morapakula et al., 2026](#); [Gong et al., 2026](#)]. In these architectures, a classical optimizer drives the quantum circuit parameters, the quantum device evaluates a cost function or generates samples, and the result is fed back classically. This division of labor plays to the strengths of both paradigms while mitigating the limitations of current quantum hardware.

The IBM Quantum platform and Qiskit framework [[Pathak et al., 2025](#); [noa](#)] have been central to financial demonstrations. Qiskit provides circuit design, simulation, error miti-

gation, and access to real hardware, making it the dominant toolkit for quantum finance research. IBM’s Qiskit Finance module includes implementations of quantum amplitude estimation for option pricing, QAOA for portfolio optimization, and quantum data loading utilities.

1.10 *Schaden’s quantum finance: a different tradition*

The term “quantum finance” predates the quantum computing era. Schaden [2002] used quantum *theory*—not quantum *computing*—to model financial markets. In Schaden’s formulation, the Hilbert space of market states consists of all possible configurations of investor holdings; the temporal evolution of an isolated market is unitary; and quantum operators represent basic financial transactions. The lognormal price distribution of classical finance emerges as the equilibrium of this quantum market model.

This tradition of applying quantum *formalism* to finance (path integrals for option pricing, Feynman–Kac formula interpretations, quantum field theory models of volatility) is distinct from the quantum *computing* approach. Both traditions use the mathematics of quantum mechanics, but for different purposes: one to model market dynamics, the other to speed up computation. This book is primarily concerned with the latter—quantum computing for finance—but the former tradition provides useful conceptual tools and is worth knowing.

1.11 *Prospects and the road ahead*

The consensus emerging from the literature is carefully optimistic. Quantum computing for finance is neither imminent revolution nor distant fantasy—it is a maturing research area with credible near-term applications and compelling long-term prospects.

The strongest near-term cases are [[Gong et al., 2026](#); [Herman et al., 2023](#); [Egger et al., 2020](#)]:

- **Hybrid optimization** for portfolio selection with cardinality or combinatorial constraints, where quantum annealing or QAOA provides meaningful advantages on NISQ hardware for moderate-size problems.
- **Quantum amplitude estimation** for Monte Carlo integration, once circuit depth and qubit count allow reliable execution of the required circuits. Error mitigation techniques are making this practical at small scales today.
- **Quantum feature transformations** for machine learning, where quantum hardware noise may actually help by providing a richer feature space than noiseless simulation—as suggested by the bond trading results of [Ciceri et al. \[2025\]](#).
- **Post-quantum security migration**, which is already underway in the financial sector and does not wait for quantum advantage.

Longer-term, fault-tolerant quantum computers will enable

the full theoretical speedups of quantum amplitude estimation (quadratic over Monte Carlo), quantum linear algebra (potentially exponential for structured linear systems), and quantum simulation of financial stochastic processes. The timeline is uncertain—perhaps 10–20 years for systems large enough to demonstrate unambiguous advantage on production-scale financial problems—but the trajectory of hardware improvement makes the direction clear.

Remark 1.2. *For finance professionals, the actionable near-term steps are: (1) build internal quantum computing expertise, (2) audit cryptographic infrastructure for post-quantum migration, (3) partner with quantum hardware providers to run proof-of-concept experiments on actual business problems, and (4) track the literature—the field is moving quickly and the competitive landscape will shift as hardware matures.*

The remaining chapters of this book develop the mathematical and algorithmic foundations needed to understand quantum finance at a technical level. We begin with the mathematical preliminaries—vector spaces, Hilbert spaces, linear operators—that underlie quantum mechanics and quantum information. From there we build toward quantum circuits, quantum algorithms, and their financial applications. The goal is to equip the reader to read the primary literature, implement quantum finance algorithms in Qiskit, and evaluate claims of quantum advantage with appropriate rigor.



Chapter 2

Mathematical Preliminaries

This chapter reviews the core mathematical structures used throughout the book. We proceed from the foundational notion of a vector space through inner products, linear operators, and spectral theory, culminating in tensor products and the Jordan canonical form. The mathematical topics covered here are essential for understanding the algebraic and geometric aspects of quantum information theory, including state spaces, quantum channels, and the behavior of quantum systems under various transformations. This is a minimal set of math I learned in the course I took on quantum and parallel computing by Alex Zecevic, as I ploughed through his book for many hours, see [Zecevic \[2022\]](#).

2.1 Vector Spaces

A *vector space* over a field $\mathbb{F}$ (typically $\mathbb{R}$ or $\mathbb{C}$) is a set V equipped with two operations—vector addition and scalar multiplication—satisfying the standard axioms of commutativity, associativity, distributivity, and the existence of an additive identity $\mathbf{0}$ and additive inverses.

Example 2.1. $\mathbb{R}^n$ with component-wise addition and scalar multiplication is the canonical real vector space. $\mathbb{C}^n$ is its complex analogue, central to quantum mechanics.

Example 2.2. The set of $m \times n$ real matrices $\mathbb{R}^{m \times n}$ forms a vector space of dimension mn .

2.2 Span and Linear Independence

Given vectors $\{v_1, \dots, v_k\} \subset V$, their *span* is

$$\text{span}\{v_1, \dots, v_k\} = \left\{ \sum_{i=1}^k \alpha_i v_i \mid \alpha_i \in \mathbb{F} \right\}.$$

The set $\{v_1, \dots, v_k\}$ is *linearly independent* if

$$\sum_{i=1}^k \alpha_i v_i = \mathbf{0} \implies \alpha_i = 0 \forall i.$$

Otherwise it is linearly dependent.

Example 2.3. In $\mathbb{R}^3$, the vectors $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ are linearly independent. Adding $(1,1,0)$ makes the set dependent since $(1,1,0) = (1,0,0) + (0,1,0)$.

2.3 Basis and Dimension

A *basis* of V is a linearly independent set that spans V . Every basis of a finite-dimensional space has the same cardinality, called the *dimension* $\dim V$.

Every vector $v \in V$ has a unique representation $v = \sum_i \alpha_i e_i$ with respect to a basis $\{e_i\}$; the scalars α_i are its *coordinates*.

Example 2.4. The standard basis of $\mathbb{C}^2$ is $\{e_1 = (1,0)^\top, e_2 = (0,1)^\top\}$. In quantum information these are denoted $|0\rangle$ and $|1\rangle$.

2.4 Inner (Scalar) Product

An *inner product* on V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ satisfying:

1. **Conjugate symmetry:** $\langle u, v \rangle = \overline{\langle v, u \rangle}$.
2. **Linearity in second argument:** $\langle u, \alpha v + \beta w \rangle = \alpha \langle u, v \rangle + \beta \langle u, w \rangle$.
3. **Positive definiteness:** $\langle v, v \rangle \geq 0$, with equality iff $v = 0$.

A vector space equipped with an inner product is called an *inner product space* (or *pre-Hilbert space*).

Example 2.5. On $\mathbb{C}^n$: $\langle u, v \rangle = u^\dagger v = \sum_{i=1}^n \bar{u}_i v_i$, where $u^\dagger$ denotes the conjugate transpose (Dirac notation: $\langle u|v \rangle$).

2.5 Norms

The inner product induces the *Euclidean* (or ℓ^2) *norm*

$$\|v\|_2 = \sqrt{\langle v, v \rangle}.$$

More generally, for $v = (v_1, \dots, v_n) \in \mathbb{R}^n$:

- **ℓ^p norm:** $\|v\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}, p \geq 1.$
- **Infinity (supremum) norm:** $\|v\|_\infty = \max_{1 \leq i \leq n} |v_i|.$
- **Weighted infinity norm:** $\|v\|_{w,\infty} = \max_{1 \leq i \leq n} \frac{|v_i|}{w_i},$ with weights $w_i > 0.$

Example 2.6. For $v = (3, -4, 1) \in \mathbb{R}^3$: $\|v\|_2 = \sqrt{26}$, $\|v\|_\infty = 4$. With weights $w = (1, 2, 1)$: $\|v\|_{w,\infty} = \max\{\frac{3}{1}, \frac{4}{2}, \frac{1}{1}\} = 3.$

2.6 Hilbert Spaces

An inner product space $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ is a *Hilbert space* if it is *complete*: every Cauchy sequence $\{v_n\} \subset \mathcal{H}$ (satisfying $\|v_m - v_n\| \rightarrow 0$ as $m, n \rightarrow \infty$) converges to a limit in $\mathcal{H}$. Completeness ensures that limits of convergent infinite series remain in the space, which is essential for spectral expansions and Fourier analysis.

The inner product induces the norm $\|v\| = \sqrt{\langle v, v \rangle}$ (Section 2.5), and the following two inequalities hold in every

inner product space:

- **Cauchy–Schwarz:** $|\langle u, v \rangle| \leq \|u\| \|v\|$, with equality iff u and v are linearly dependent.
- **Parallelogram law:** $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$.

A Hilbert space is *separable* if it admits a countable orthonormal basis $\{e_i\}$ such that every $v \in \mathcal{H}$ has the expansion $v = \sum_i \langle e_i, v \rangle e_i$. All Hilbert spaces arising in quantum mechanics and this book are separable. The *Riesz representation theorem* states that every bounded linear functional $\varphi : \mathcal{H} \rightarrow \mathbb{F}$ has the form $\varphi(v) = \langle u, v \rangle$ for a unique $u \in \mathcal{H}$ (in Dirac notation: every bra $\langle u|$ corresponds to a unique ket $|u\rangle$).

Example 2.7. $\mathbb{C}^n$ with inner product $\langle u, v \rangle = u^\dagger v$ is a finite-dimensional Hilbert space; completeness is automatic in finite dimensions. The state space of an n -level quantum system is $\mathbb{C}^n$.

Example 2.8. $L^2[a, b]$ (square-integrable functions with $\langle f, g \rangle = \int_a^b f^*(x) g(x) dx$) is an infinite-dimensional separable Hilbert space. It is the natural setting for wave functions and Fourier analysis; see Section 2.22.

2.7 Orthogonality and Orthonormal Bases

Two vectors u, v are *orthogonal*, written $u \perp v$, if $\langle u, v \rangle = 0$. A set $\{e_1, \dots, e_n\}$ is *orthonormal* if

$$\langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

An orthonormal basis (ONB) simplifies coordinate extraction: $\alpha_i = \langle e_i, v \rangle$.

Example 2.9. The standard basis $\{e_i\}$ of $\mathbb{R}^n$ is orthonormal under the Euclidean inner product. The Bell basis $\{|\Phi^\pm\rangle, |\Psi^\pm\rangle\}$ is an ONB for $\mathbb{C}^4$ used in quantum information.

2.8 Gram–Schmidt Orthogonalization

Given a linearly independent set $\{v_1, \dots, v_k\}$, the *Gram–Schmidt process* produces an orthonormal set $\{e_1, \dots, e_k\}$ spanning the same subspace:

$$u_j = v_j - \sum_{i=1}^{j-1} \langle e_i, v_j \rangle e_i, \quad e_j = \frac{u_j}{\|u_j\|}.$$

Example 2.10. Starting from $v_1 = (1, 1, 0)^\top$, $v_2 = (1, 0, 1)^\top$ in $\mathbb{R}^3$:

$$e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}}(1, 1, 0)^\top, \quad u_2 = v_2 - \langle e_1, v_2 \rangle e_1 = \frac{1}{2}(1, -1, 2)^\top.$$

2.9 Linear Operators

A map $A : V \rightarrow W$ is a *linear operator* if $A(\alpha u + \beta v) = \alpha Au + \beta Av$. When $V = W$ we call A a *linear operator on V* . With respect to bases, A is represented by a matrix; composition of operators corresponds to matrix multiplication.

Key properties:

- **Kernel:** $\ker A = \{v \in V \mid Av = \mathbf{0}\}$.
- **Image:** $\operatorname{im} A = \{Av \mid v \in V\}$.
- **Rank-nullity theorem:** $\dim \ker A + \dim \operatorname{im} A = \dim V$.

2.10 Eigenvalues and Eigenvectors

A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of A if there exists a nonzero $v \in V$ such that

$$Av = \lambda v.$$

Such v is an *eigenvector*. Eigenvalues are roots of the *characteristic polynomial* $p(\lambda) = \det(A - \lambda I)$.

Example 2.11. For $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, $p(\lambda) = (\lambda - 2)(\lambda - 3)$, giving eigenvalues $\lambda_1 = 2$, $\lambda_2 = 3$ with eigenvectors $(1, 0)^\top$ and $(1, 1)^\top$ respectively.

2.11 Nonsingular Matrices

A square matrix $A \in \mathbb{F}^{n \times n}$ is *nonsingular* (invertible) if any of the following equivalent conditions hold:

- $\det A \neq 0$.
- $\ker A = \{\mathbf{0}\}$ (zero is not an eigenvalue).
- A has full rank n .
- There exists A^{-1} such that $AA^{-1} = A^{-1}A = I$.

2.12 Similar Matrices

Matrices A and B are *similar* if there exists a nonsingular P such that $B = P^{-1}AP$. Similar matrices represent the same linear operator in different bases and share eigenvalues, determinant, trace, and characteristic polynomial.

Example 2.12. If A is diagonalizable with eigenvalues λ_i and eigenvector matrix P , then $P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n)$.

2.13 Hermitian Operators

An operator A on an inner product space is *Hermitian* (self-adjoint) if $A = A^\dagger$, i.e., $\langle u, Av \rangle = \langle Au, v \rangle$ for all u, v . Key properties:

- All eigenvalues are real.

- Eigenvectors for distinct eigenvalues are orthogonal.
- Every Hermitian operator is diagonalizable by a unitary matrix.

Example 2.13. The Pauli matrices $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ are Hermitian with eigenvalues ± 1 .

2.14 Unitary Operators

An operator U is *unitary* if $U^\dagger U = UU^\dagger = I$, equivalently if it preserves inner products: $\langle Uu, Uv \rangle = \langle u, v \rangle$. Unitary operators map ONBs to ONBs, have $|\det U| = 1$, and all eigenvalues lie on the unit circle in $\mathbb{C}$.

Example 2.14. The Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is unitary and Hermitian ($H = H^\dagger = H^{-1}$), and maps the computational basis to an equal superposition.

2.15 Spectral Decomposition

The *spectral theorem* states that every Hermitian (or, over $\mathbb{C}$, every normal: $AA^\dagger = A^\dagger A$) operator A admits the decomposition

$$A = \sum_i \lambda_i |e_i\rangle \langle e_i| = \sum_i \lambda_i P_i,$$

where λ_i are eigenvalues, $|e_i\rangle$ are corresponding orthonormal eigenvectors, and $P_i = |e_i\rangle\langle e_i|$ are rank-1 orthogonal projectors satisfying $P_i P_j = \delta_{ij} P_i$ and $\sum_i P_i = I$.

This decomposition allows the definition of functions of operators: $f(A) = \sum_i f(\lambda_i) P_i$.

Example 2.15. For $A = \sigma_z$ with eigenvalues ± 1 and eigenvectors $|0\rangle, |1\rangle$:

$$\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|, \quad e^{i\theta\sigma_z} = e^{i\theta}|0\rangle\langle 0| + e^{-i\theta}|1\rangle\langle 1|.$$

2.16 Tensor Product of Vector Spaces

Given vector spaces V (dim m) and W (dim n), their *tensor product* $V \otimes W$ is a vector space of dimension mn whose elements are (equivalence classes of) formal sums of *simple tensors* $v \otimes w$, with bilinearity:

$$(\alpha v) \otimes w = v \otimes (\alpha w) = \alpha(v \otimes w), \quad (v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w.$$

If $\{e_i\}$ is a basis of V and $\{f_j\}$ is a basis of W , then $\{e_i \otimes f_j\}$ is a basis of $V \otimes W$.

Example 2.16. For $\mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$, the computational basis is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ where $|ij\rangle = |i\rangle \otimes |j\rangle$. An entangled state such as $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ cannot be written as a simple tensor $v \otimes w$.

2.17 Tensor Product of Operators

Given $A : V \rightarrow V$ and $B : W \rightarrow W$, their tensor product $A \otimes B : V \otimes W \rightarrow V \otimes W$ is defined by

$$(A \otimes B)(v \otimes w) = (Av) \otimes (Bw),$$

extended by linearity. In matrix form, $A \otimes B$ is the *Kronecker product*: if $A = (a_{ij})$, then

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}.$$

Example 2.17. $\sigma_z \otimes I_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$ measures the z -spin of the first qubit in a two-qubit system.

2.18 Jordan Canonical Form

Not every matrix is diagonalizable. Every $A \in \mathbb{C}^{n \times n}$ is similar to its *Jordan canonical form*

$$J = P^{-1}AP = \text{diag}(J_{n_1}(\lambda_1), \dots, J_{n_k}(\lambda_k)),$$

where each *Jordan block* is

$$J_m(\lambda) = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix} \in \mathbb{C}^{m \times m}.$$

The number and sizes of Jordan blocks for eigenvalue λ are determined by the dimensions of $\ker(A - \lambda I)^k$ for $k = 1, 2, \dots$. The Jordan form is unique up to reordering of blocks.

Example 2.18. The matrix $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ is already in Jordan form with a single 2×2 block $J_2(3)$. It has eigenvalue $\lambda = 3$ (algebraic multiplicity 2, geometric multiplicity 1) and is not diagonalizable.

The Jordan form is essential for computing matrix exponentials and analyzing the long-run behavior of non-diagonalizable systems, relevant in the study of quantum channels and decoherence.

2.19 Matrix Norms

Every matrix $A \in \mathbb{F}^{m \times n}$ induces an *operator norm* $\|A\|_{\text{op}} = \max_{\|x\|=1} \|Ax\|$. The induced norms most relevant here are:

- **Spectral (2-)norm:** $\|A\|_2 = \sqrt{\rho(A^\top A)}$, where $\rho(A^\top A) = \max_i \lambda_i(A^\top A)$ is the *spectral radius* of $A^\top A$.

• **Induced infinity norm:** $\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$ (maximum absolute row sum).

• **Weighted infinity norm:** $\|A\|_{\omega, \infty} = \max_{1 \leq i \leq m} \frac{1}{\omega_i} \sum_{j=1}^n |a_{ij}| \omega_j$,
with weights $\omega_j > 0$.

All matrix norms satisfy $\|A\| \geq 0$, $\|cA\| = |c|\|A\|$, the triangle inequality, and submultiplicativity $\|AB\| \leq \|A\|\|B\|$.

Example 2.19. For $A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 4 & 5 \\ 1 & 2 & 3 \end{pmatrix}$, the eigenvalues of $A^\top A$ are approximately 0.310, 13.776, 45.914, giving $\|A\|_2 = \sqrt{45.914} \approx 6.776$. The induced infinity norm is $\|A\|_\infty = \max\{3, 9, 6\} = 9$. With weights $\omega = (1, 0.5, 0.4)$:

$$\|A\|_{\omega, \infty} = \max\left\{\frac{1 \cdot 1 + 0 \cdot 0.5 + 2 \cdot 0.4}{1}, \frac{0 \cdot 1 + 4 \cdot 0.5 + 5 \cdot 0.4}{0.5}, \frac{1 \cdot 1 + 2 \cdot 0.5 + 3 \cdot 0.4}{0.4}\right\}$$

2.20 Quadratic Forms

Given a symmetric matrix $A \in \mathbb{R}^{n \times n}$, the associated *quadratic form* is $q(x) = x^\top Ax$. The matrix A is *positive definite* if $q(x) > 0$ for all $x \neq \mathbf{0}$, and *positive semi-definite* if $q(x) \geq 0$ for all x .

Via the spectral decomposition $A = T\Lambda T^\top$ (with T ortho-

nal), the substitution $y = T^\top x$ gives

$$q(x) = x^\top Ax = y^\top \Lambda y = \sum_{i=1}^n \lambda_i y_i^2.$$

Hence A is positive definite if and only if all eigenvalues $\lambda_i > 0$. Choosing $x = x_k$ (the k -th eigenvector) gives $q(x_k) = \lambda_k \|x_k\|_2^2$, so any $\lambda_k \leq 0$ immediately violates positive definiteness.

Example 2.20. $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ has characteristic polynomial $\lambda^2 - 5\lambda + 5 = 0$, giving $\lambda_{1,2} = \frac{5 \pm \sqrt{5}}{2} \approx 1.382, 3.618$. Both are positive, confirming $q(x) = 2x_1^2 + 2x_1x_2 + 3x_2^2 > 0$ for all $x \neq 0$.

2.21 Functions of Matrices

For f with Taylor series $f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$, the *matrix function* $f(A)$ is defined by

$$f(A) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} A^k.$$

For a diagonalizable $A = T\Lambda T^{-1}$, this simplifies to

$$f(A) = T f(\Lambda) T^{-1}, \quad f(\Lambda) = \text{diag}(f(\lambda_1), \dots, f(\lambda_n)),$$

since $A^k x_i = \lambda_i^k x_i$ implies $f(A)x_i = f(\lambda_i)x_i$ for each eigenvector x_i . For a Hermitian $A = \sum_i \lambda_i P_i$, the spectral form is $f(A) = \sum_i f(\lambda_i) P_i$, consistent with Section 2.15.

Example 2.21. Let $A = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$ and $f(x) = e^{-x} \cos x$. The eigenvalues are $\lambda_{1,2} = 3 \mp \sqrt{5}$ ($\approx 0.764, 5.236$). Computing:

$$f(\Lambda) \approx \begin{pmatrix} 0.336 & 0 \\ 0 & 0.003 \end{pmatrix}, \quad f(A) = T f(\Lambda) T^{-1} \approx \begin{pmatrix} 0.244 & \\ & 0.149 \end{pmatrix}$$

2.22 L^2 Function Spaces

The space $L^2[a, b]$ consists of complex-valued functions $f: [a, b] \rightarrow \mathbb{C}$ that are square-integrable, $\int_a^b |f(x)|^2 dx < \infty$. It is an infinite-dimensional *Hilbert space* with inner product

$$\langle f, g \rangle = \int_a^b f^*(x) g(x) dx,$$

satisfying the same axioms as the finite-dimensional case: conjugate symmetry $\langle f, g \rangle = \overline{\langle g, f \rangle}$, linearity in the second argument, and positive definiteness. A countable set $\{\phi_i\}$ is an *orthonormal system* if $\langle \phi_i, \phi_j \rangle = \delta_{ij}$. When it spans $L^2[a, b]$, every f has the expansion

$$f = \sum_{i=1}^{\infty} \langle \phi_i, f \rangle \phi_i,$$

the infinite-dimensional analogue of coordinate extraction via an ONB.

2.23 Fourier Series

A T -periodic function satisfies $f(t+T) = f(t)$. With $\omega = 2\pi/T$, the *Fourier series* is

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t).$$

The functions $\{1, \cos(n\omega t), \sin(n\omega t)\}_{n \geq 1}$ are orthogonal in $L^2[0, T]$, giving

$$a_0 = \frac{1}{T} \int_0^T f(t) dt, \quad a_n = \frac{2}{T} \int_0^T f(t) \cos(n\omega t) dt, \quad b_n = \frac{2}{T} \int_0^T f(t) \sin(n\omega t) dt.$$

For even functions ($f(t) = f(-t)$), all $b_n = 0$. The Fourier basis plays the same structural role in $L^2[0, T]$ as the monomial basis $\{1, x, x^2, \dots\}$ in a Taylor expansion, but orthogonality makes coefficient extraction direct.

2.24 Eigenfunctions

A linear operator $\hat{A}$ on a Hilbert space has an *eigenfunction* ϕ_i with *eigenvalue* λ_i if $\hat{A}\phi_i = \lambda_i\phi_i$. When $\hat{A}$ possesses an orthonormal set of eigenfunctions $\{\phi_i\}$ spanning the space, the action on any $f = \sum_i \beta_i \phi_i$ is computed term-by-term:

$$\hat{A}f = \sum_i \lambda_i \beta_i \phi_i, \quad \beta_i = \langle \phi_i, f \rangle.$$

Example 2.22. Let $\{\psi_0, \psi_1\}$ be an orthonormal basis and define the bit-flip operator $\hat{X}$ by $\hat{X}\psi_0 = \psi_1$, $\hat{X}\psi_1 = \psi_0$. For

$\psi = \alpha_0 \psi_0 + \alpha_1 \psi_1$, the eigenfunction equation $\hat{X}\psi = \lambda\psi$ gives $\lambda^2 = 1$, so $\lambda_1 = 1$ and $\lambda_2 = -1$. The normalized eigenfunctions are

$$\psi_A = \frac{1}{\sqrt{2}}(\psi_0 + \psi_1), \quad \psi_B = \frac{1}{\sqrt{2}}(\psi_0 - \psi_1).$$

2.25 Hadamard Operator

The Hadamard operator $\hat{H}$ acts on the orthonormal basis $\{\psi_0, \psi_1\}$ by

$$\hat{H}\psi_0 = \frac{1}{\sqrt{2}}(\psi_0 + \psi_1), \quad \hat{H}\psi_1 = \frac{1}{\sqrt{2}}(\psi_0 - \psi_1).$$

It is both self-adjoint and unitary ($\hat{H} = \hat{H}^\dagger = \hat{H}^{-1}$). The characteristic equation is $\lambda^2 - 1 = 0$, giving $\lambda_1 = 1$ and $\lambda_2 = -1$, with normalized eigenfunctions

$$\psi_A = \cos \frac{\pi}{8} \psi_0 + \sin \frac{\pi}{8} \psi_1 \approx 0.9239 \psi_0 + 0.3827 \psi_1, \quad \psi_B = -\sin \frac{\pi}{8} \psi_0 + \cos \frac{\pi}{8} \psi_1 \approx -0.3827 \psi_0 + 0.9239 \psi_1$$

2.26 Adjoint, Self-Adjoint, and Unitary Operators on Function Spaces

For a linear operator $\hat{A}$ on a Hilbert space $\mathcal{H}$, the adjoint $\hat{A}^\dagger$ is the unique operator satisfying

$$\langle \hat{A}f, g \rangle = \langle f, \hat{A}^\dagger g \rangle \quad \forall f, g \in \mathcal{H}.$$

- $\hat{A}$ is *self-adjoint* (Hermitian) if $\hat{A} = \hat{A}^\dagger$: eigenvalues are real and eigenfunctions for distinct eigenvalues are orthogonal.
- $\hat{U}$ is *unitary* if $\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{I}$, equivalently $\langle \hat{U}f, \hat{U}g \rangle = \langle f, g \rangle$. All eigenvalues satisfy $|\lambda_k| = 1$.

The unit-circle property follows directly: if $\hat{U}\xi_k = \lambda_k \xi_k$, then

$$\langle \xi_k, \xi_k \rangle = \langle \hat{U}\xi_k, \hat{U}\xi_k \rangle = |\lambda_k|^2 \langle \xi_k, \xi_k \rangle \implies |\lambda_k| = 1.$$

These properties mirror exactly those of Hermitian and unitary matrices, confirming that the matrix and functional-space theories are instances of the same abstract framework.

2.27 Matrix Representations of Operators

Given operator $\hat{A}$ on a space with orthonormal basis $\{\psi_j\}_{j=0}^{n-1}$, its *matrix representation* A has entries

$$A_{jk} = \langle \psi_j, \hat{A}\psi_k \rangle,$$

so that $\hat{A}\psi_k = \sum_j A_{jk}\psi_j$. The coordinates transform as $\rho = A\alpha$: if $f = \sum_k \alpha_k \psi_k$ then $\hat{A}f = \sum_j \rho_j \psi_j$ with $\rho_j = \sum_k A_{jk}\alpha_k$.

Example 2.23. In basis $\{\psi_0, \psi_1\}$, the operators $\hat{X}$ and $\hat{H}$ have matrix representations

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

matching the Pauli matrix σ_x and the Hadamard gate from Sections 2.13 and 2.14.

2.28 Coordinate Transformations

Let $\{\psi_j\}$ and $\{\xi_k\}$ be two orthonormal bases. The *change-of-basis matrix* S with entries

$$S_{jk} = \langle \psi_j, \xi_k \rangle$$

encodes the ψ -coordinates of each new basis vector as columns. If f has coordinates α in the ψ -basis and β in the ξ -basis, then $\alpha = S\beta$. Since both bases are orthonormal, S is unitary ($S^{-1} = S^\dagger$), and the matrix representation of $\hat{A}$ transforms as

$$A_\xi = S^{-1} A_\psi S = S^\dagger A_\psi S,$$

making A_ξ and A_ψ similar matrices with identical eigenvalues.

Example 2.24. *The Hadamard operator has matrix $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ in the $\{\psi_0, \psi_1\}$ basis. In its eigenbasis $\{\psi_A, \psi_B\}$ the representation is $H_{\text{eig}} = \text{diag}(1, -1)$. The change-of-basis matrix S with columns equal to the ψ -coordinates of ψ_A and ψ_B satisfies $S^\dagger H S = \text{diag}(1, -1)$:*

$$S = \begin{pmatrix} \cos \frac{\pi}{8} & -\sin \frac{\pi}{8} \\ \sin \frac{\pi}{8} & \cos \frac{\pi}{8} \end{pmatrix} \approx \begin{pmatrix} 0.9239 & -0.3827 \\ 0.3827 & 0.9239 \end{pmatrix}.$$

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